



香 港 大 學

THE UNIVERSITY OF HONG KONG

**MSc in Computer Science -
COMP 7705 Capstone Project**

Interim Report

A Multimodal AI-Based Mobile Application for
Intelligent Household Maintenance Diagnostics
and Management

Group Members:

MOK Yuen Man 3036383381

KWAN Kai Yin 3036412182

LAI Long Wah 3036383276

Project Mentor:

Dr. CHIM Tat Wing

Table of Contents

1. Introduction.....	- 1 -
2. Literature Review on AI Technologies for Home Maintenance.....	- 2 -
3. Project Motivation	- 3 -
4. Project Aims and Objectives.....	- 4 -
5. Overview and Scope of the Mobile Application	- 4 -
6. System Architecture and Technical Details.....	- 5 -
6.1 Architecture Overview	- 5 -
6.2 Mobile Application Frontend (iOS — SwiftUI)	- 6 -
6.3 Backend and API Layer (FastAPI)	- 11 -
6.4 AI Components and Multimodal RAG Pipeline	- 12 -
6.5 Data Storage	- 14 -
6.6 Deployment and Security	- 15 -
7. Preliminary Results	- 15 -
7.1 Frontend Prototype (UI/UX Design)	- 15 -
7.2 Backend Prototype (AI Pipeline).....	- 16 -
8. Project Status and Remaining Schedule.....	- 21 -
9. References.....	- 23 -

1. Introduction

Residential environments in Hong Kong rely on tightly interconnected systems, including electrical wiring, plumbing networks, home appliances, and structural components. Maintaining these systems in a timely and informed manner is essential for household safety, operational efficiency, and long-term cost control. A research published in 2025 shows that planned, proactive maintenance can reduce long-term costs, prevent escalation of minor faults, and extend the service lifespan of household assets (Decades Construction Group, 2025). However, in practice, household maintenance in Hong Kong remains largely reactive and unstructured, leaving homeowners vulnerable to unexpected failures and avoidable expenses.

A key barrier to effective household maintenance is economic. In Hong Kong, popular appliance brands such as Hitachi and Siemens usually charge a non-refundable inspection fee, ranging from HK\$460 to HK\$850, before any repair work begins, even if the product cannot successfully be repaired (Arçelik Hitachi Home Appliances Sales Hong Kong Limited, 2024; BSH Home Appliances Limited, 2024). A report published by the Business Environment Council further stated that consumers frequently choose to replace or dispose of home appliances due to complicated maintenance procedures and disproportionately high repair costs relative to replacement cost. As a result, replacement is often perceived as more cost-effective than repair, contributing to approximately 70,000 tonnes of e-waste generated in Hong Kong each year, alongside 1,800 tonnes of furniture discarded daily, accounting for nearly 16% of municipal solid waste by weight. A considerable proportion of this waste could have been avoided through timely maintenance and repair, aligning with the broader source reduction and recovery goals set out in Hong Kong's *Waste Blueprint for Hong Kong 2035* (Business Environment Council, 2023).

In addition to cost barriers, information asymmetry presents an additional challenge. A study by the Consumer Council in 2023 found that most homeowners lack professional knowledge to assess fault severity or evaluate whether a repair quotation is reasonable, making them unable to distinguish between minor and major faults, or determine when professional assistance is truly necessary, and must instead rely on the judgment and recommendations of technicians (Consumer Council, 2023). This dependence enables exploitative practices, including inflated quotations, unnecessary work recommendations, and a lack of transparency over what they are paying for. In 2024, the Consumer Council recorded 1,359 complaints related to repair and maintenance services, of which 45% involved electrical appliances, reflecting the scale and prevalence of such malpractices (Consumer Council, 2025). Moreover, separate 2024 study on the home renovation industry also found that a fifth of consumers had disputes with appointed contractors, most commonly over defective work, delays, and failure to fix the identified faults (Consumer Council, 2024). These figures indicate systemic quality and transparency deficiencies in Hong Kong's home maintenance sector.

2. Literature Review on AI Technologies for Home Maintenance

Recent advances in artificial intelligence offer a technical foundation to address the barriers identified above. Vision-language models (VLMs) have emerged as a cornerstone for image-based fault identification. Kumar et al. (2024) developed Diagnostics-LLaVA, a domain-specific VLM built on the LLaVA architecture, which excels at interpreting equipment images for fault diagnostics, root-cause analysis, and repair recommendations in Prognosis and Health Management (PHM) tasks.

Multimodal generative AI assistants further extend these capabilities. Systems built with Amazon Bedrock (using Claude 3 models) support predictive maintenance and root-cause diagnosis by processing multiple modalities—including user-uploaded images of damaged components, PDF manuals, videos, and audio — through multimodal retrieval-augmented generation, delivering faster and more accurate troubleshooting for technicians and users alike (Amazon Web Services, 2025).

To mitigate hallucinations and ensure responses are grounded in reliable sources, Retrieval-Augmented Generation (RAG) frameworks have been widely adopted for technical documentation. Practical implementations, such as Elasticsearch-based RAG pipelines, ingest appliance manuals (PDFs split and embedded via models like sentence-transformers), perform semantic search on user queries, and synthesize accurate, context-aware answers — directly applicable to household troubleshooting scenarios like resetting dishwashers or identifying error codes (Elasticsearch, 2025).

Consumer-facing tools demonstrate real-world viability. iFixit’s FixBot (released December 2025) combines computer vision for photo-based device identification and component analysis with a custom retrieval engine drawing from 125,000+ repair guides and manuals, enabling conversational, step-by-step guidance via mobile camera for appliances (e.g., washing machines, refrigerators) and other household items (iFixit, 2025). Complementary mobile applications, such as *HomeCare Manager* and *HomeKeep*, incorporate AI for maintenance scheduling, reminders, and basic asset tracking.

Despite these developments, notable gaps persist. No unified consumer mobile application currently integrates fault diagnostics, RAG-grounded repair guidance, safety classification, and appliance lifecycle management into a single safety-conscious system tailored for household use. This gap represents a meaningful opportunity for a consumer-focused solution that consolidates these capabilities into a secure, deployable platform — an opportunity this project directly addresses.

3. Project Motivation

Despite the availability of professional repair services in Hong Kong, many common household issues are minor and do not necessarily require immediate expert intervention. Problems such as leaking taps, clogged drains, tripped circuit breakers, or appliance error codes can often be resolved through basic troubleshooting, provided that clear instructions and appropriate safety guidance are available. However, in practice, homeowners are rarely equipped with the knowledge required to confidently assess fault severity or determine whether professional assistance is truly necessary.

This lack of technical understanding creates a strong dependence on external technicians, exposing consumers to the exploitative practices documented above — inflated quotations, unnecessary repair recommendations, and limited pricing transparency. Without the ability to independently verify such claims, homeowners are left with limited negotiating power and often incur avoidable expenses. Consequently, even manageable issues are frequently escalated into expensive service calls or premature appliance replacements.

In addition to diagnostic uncertainty, household maintenance is further complicated by fragmented information management. Appliance manuals, warranty records, and maintenance histories are often dispersed across different physical and digital formats, making them difficult to retrieve when needed. Furthermore, these materials are typically written in technical language that is not easily understood by non-experts. The absence of an integrated system for organizing, accessing, and interpreting such information significantly reduces a homeowner's ability to make informed maintenance decisions.

Recent advancements in multimodal artificial intelligence present a compelling opportunity to address these challenges. Vision-Language Models (VLMs) enable intuitive, image-based fault identification, allowing users to interact with systems using photos rather than technical descriptions. At the same time, Retrieval-Augmented Generation (RAG) frameworks ensure that responses are grounded in authoritative sources such as manufacturer manuals and regulatory guidelines, thereby improving both accuracy and trustworthiness. When combined with structured safety mechanisms, these technologies can provide not only diagnostic insights but also locally grounded guidance — aligned with Hong Kong regulatory standards — on appropriate next steps.

Motivated by these gaps, this project seeks to develop a unified, AI-powered mobile application that supports informed decision-making in household maintenance. Rather than functioning solely as a record-keeping or reminder tool, the proposed system emphasizes AI-assisted diagnosis, safety-aware guidance, and contextualized recommendations prior to any maintenance action being taken. By integrating fault detection, evidence-based troubleshooting, risk classification, and appliance lifecycle management into a single platform, the application aims to reduce information asymmetry, minimize unnecessary repair costs, and promote proactive, sustainable maintenance practices among homeowners in Hong Kong.

4. Project Aims and Objectives

This project aims to design and implement a multimodal AI-powered mobile application capable of household issues diagnostics, grounded repair guidance, and centralizing appliance maintenance management by:

- i. Integrating Vision-Language Model (VLM) diagnostic pipeline for appliance identification and visual fault/error-code extraction.
- ii. Implementing a Retrieval-Augmented Generation (RAG) framework using a curated, Hong Kong-specific knowledge base of official manufacturer manuals and Electrical and the Mechanical Services Department (EMSD)/GovHK regulatory guidance.
- iii. Designing a deterministic Safety Guardrail and Severity Classification mechanism (Low / Medium / High) with hard-coded disclaimers and referral protocols for high-risk scenarios.
- iv. Building an iOS-native appliance asset management module for warranty tracking and rule-based maintenance scheduling.
- v. Evaluating the system using structured benchmarks covering retrieval precision, LLM faithfulness, and VLM diagnostic accuracy.

5. Overview and Scope of the Mobile Application

UncleFix – AI-Powered Home Repair Assistant, is a multimodal mobile application for fault diagnosis, guided repairs, appliance tracking, and maintenance reminders.

Category: Utilities

Target Audience: UncleFix is designed for Hong Kong residents who lack technical expertise in household damages, defects, and appliance repair, and seek reliable, AI-powered guidance to diagnose issues and make informed maintenance decisions.

To ensure technical feasibility, safety compliance, and data grounding density, the operational scope of the proposed prototype is explicitly defined along four dimensions:

- i. **In-Scope Domestic Tasks:** The application focuses specifically on minor interior household issues that fall within the safe scope of homeowner self-assessment. This primarily includes common kitchen and laundry appliance faults (e.g., washing machines, refrigerators, microwave ovens, etc.), minor plumbing issues like leaking taps or clogged drains, and basic electrical resets such as identifying tripped circuit breakers.
- ii. **Out-of-Scope Exceptions:** Structural repairs, major plumbing overhauls, gas-line configurations, or building-wide infrastructural issues are strictly out of scope. For hazardous or legally regulated interventions, the prototype acts strictly as a hazard-warning and referral mechanism rather than a DIY guide.

- iii. **System Outputs:** For every diagnostic request, the application is bounded to produce the following structured output:
 - The likely fault cause and category of the fault;
 - An estimated fixing time and a deterministic safety risk rating (Low / Moderate / High);
 - A list of tools and materials required; and
 - A recommended course of action (Independent DIY, ongoing monitoring, or immediate professional intervention) with safety notes.
- iv. **Geographic and Regulatory Localization:** The RAG knowledge base is exclusively curated from Hong Kong-specific regulatory and consumer sources. This includes technical guidance from EMSD, Consumer Council appliance reliability data, GovHK household maintenance advisories, and manufacturer’s user manual uploaded by users, ensuring outputs strictly comply with local electrical standards and regional housing configurations.

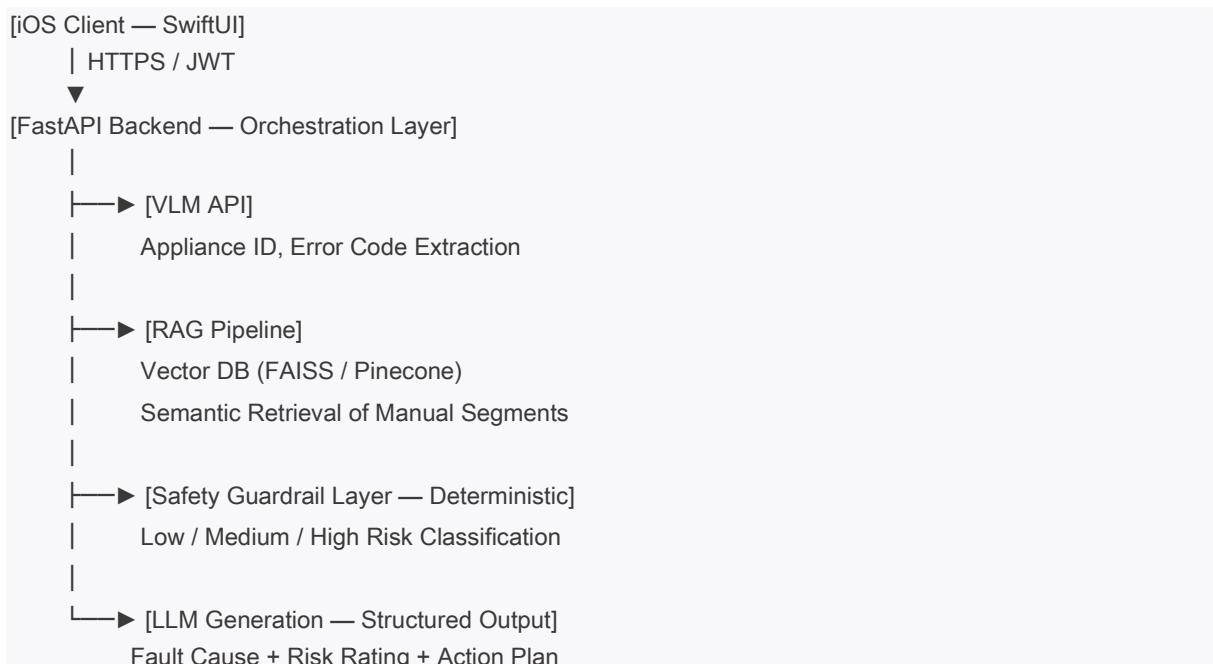
6. System Architecture and Technical Details

6.1 Architecture Overview

The mobile application follows a layered modular architecture comprising four primary layers:

- i. the iOS Presentation Layer,
- ii. the Application and API Layer (FastAPI),
- iii. the AI and Retrieval Layer, and
- iv. the Data Storage Layer.

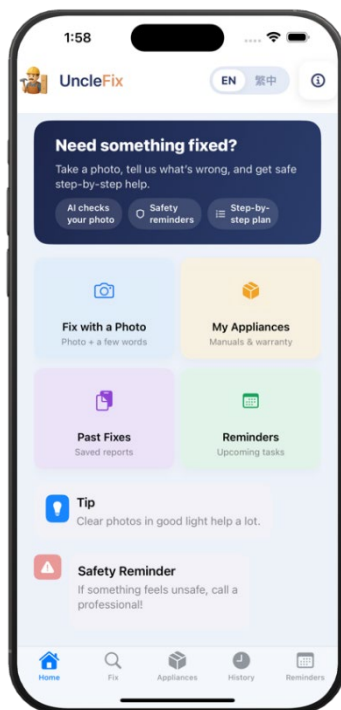
The architecture prioritizes modularity, safety enforcement, and local regulatory compliance.



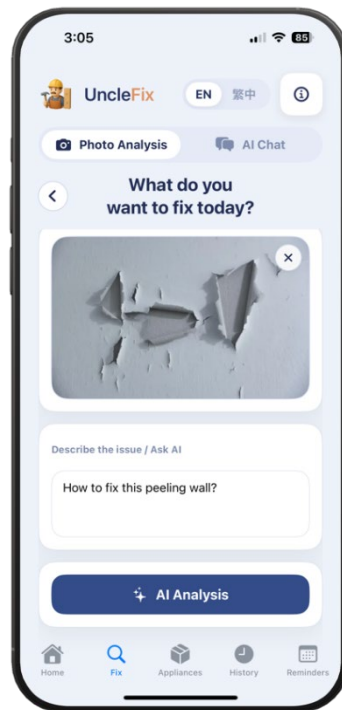
6.2 Mobile Application Frontend (iOS — SwiftUI)

6.2.1 Screen Design and User Interaction Flow

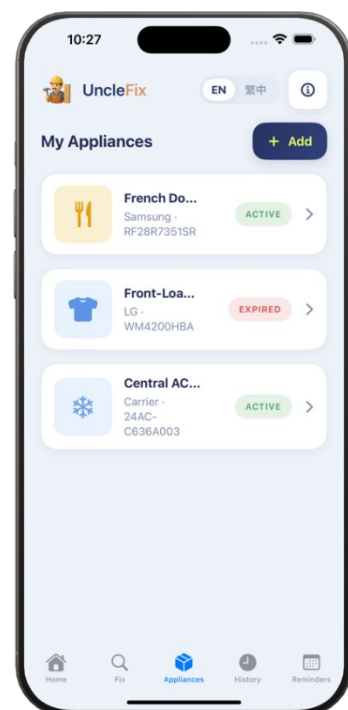
The frontend of the mobile application was developed using Swift and SwiftUI, targeting iOS 17+. The Xcode project implements the following five core screens:



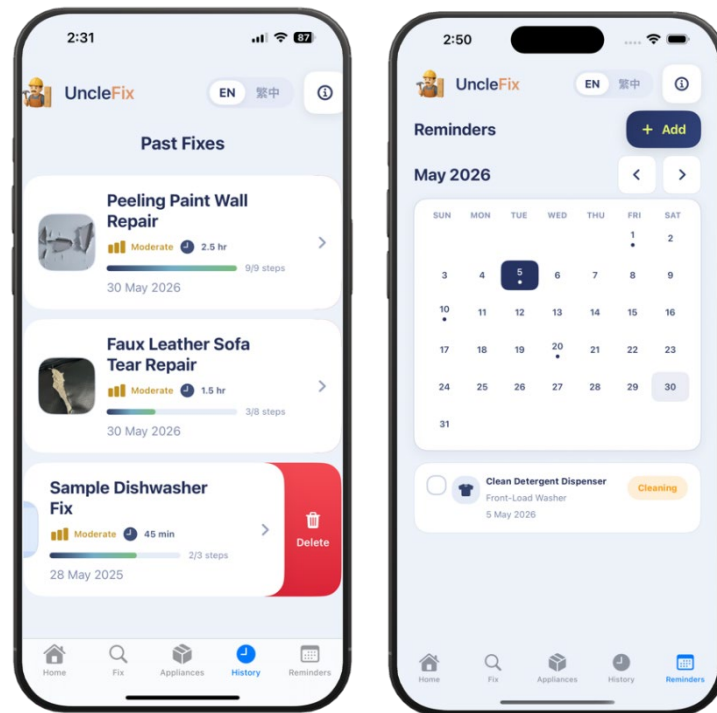
Home



Fix



Appliances



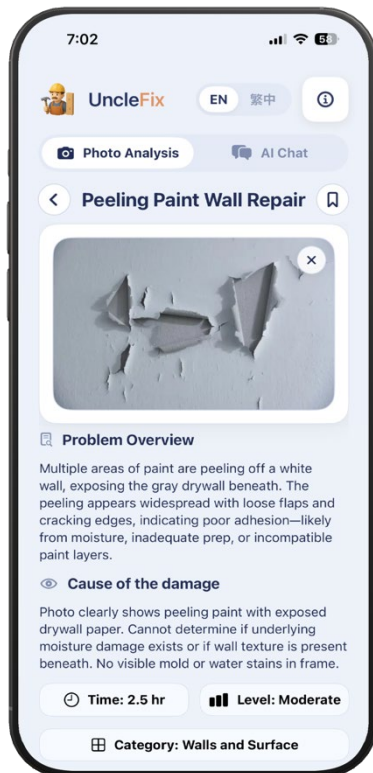
History

Reminders

Screen	Functionality
Home	Central navigation hub providing one-tap access to all core screens – Fix, Appliances, History, Reminders. A safety reminder "If something feels unsafe, call a professional!" is persistently displayed on the screen, consistent with the Safety-First design philosophy
Fix	AI-powered diagnostic interface for fault identification and repair guidance via photo taking/upload and issue description
Appliances	Appliance asset management for registering devices, uploading manuals, tracking warranties, and setting up maintenance tasks
History	Chronological record of past maintenance activities and AI-generated repair recommendations
Reminders	Manage scheduled maintenance with notifications

The following subsections provide a detailed walkthrough of the key inner screens, illustrating the user interaction flow and the underlying functionality of the specific screens. Each screen is designed with a focus on simplicity and accessibility, ensuring that users can navigate the mobile application and interpret diagnostic results with confidence.

AI Diagnostic Response on the *Fix Screen*



Upon submitting a photo and issue description, the screen displays the AI-generated diagnostic result structured into three components: estimated fixing time and severity rating (Low / Moderate / High), a repair guidance with step-by-step recommended actions with repair tips, and a safety disclaimer. All responses are grounded in the knowledge base to ensure accuracy and local regulatory compliance.

Appliance Registration on the *Appliances Screen*

4:01

Add Appliance

APPLIANCE NAME

Air Fryer

CATEGORY

Kitchen

BRAND / MANUFACTURER

PHILIPS

MODEL NUMBER

NA231/09 2000

PURCHASE DATE

27 May 2026

WARRANTY EXPIRY

26 May 2028

NOTES

Volume: 6.2L

Temperature: 60-200°C

ADD PHOTO

Take Photo

Add Photo

USER MANUAL (PDF OR TEXT FILE)

Airfryer_Manual.pdf

Upload Manual

TASK

Deep oil cleaning

Schedule

30 May 2026

Cleaning

Save Appliance

The appliance registration allows users to add a new household appliance by entering details including the category, brand, model number, purchase date, and warranty expiry date. Users may also add maintenance tasks as well as upload the manufacturer's user manual in PDF/TEXT format, which is stored for reference and indexed into the personal knowledge base for future diagnostic queries.

- 9 -

Appliance Record on the *Appliances Screen*

Energy Saving & Maintenance Tips
AI-powered tips for this appliance

Get personalised energy-saving and e...

- The Philips NA231/09 is a 2000W dual-basket air fryer—its high wattage demands smart scheduling to avoid peak electricity rates.
- Dual-basket design lets you cook foods separately; mismatched timing wastes energy on reheating one basket.

Hide details

Tips

- Use "Sync Finish" to align both baskets' end times, avoiding idle heating of finished food.
- Fill baskets to recommended capacity (not overcrowded, not underfilled); empty space still heats inefficiently.

Maintenance

- Clean heating element monthly with soft brush—carbon buildup forces longer cook times.
- Check basket seals for food debris; gaps cause heat leakage and extended cycles.
- Descale if hard water affects any steam functions (if applicable to your variant).

RELATED TASKS

DEEP OIL CLEANING
30 May 2026

Edit Delete Ask

The appliance record displays a summary of the registered appliances, showing each device's name, warranty status, and upcoming maintenance tasks. A colour-coded warranty indicator alerts users when a warranty is approaching expiry or has expired, enabling timely action before repair costs become the user's sole responsibility. Energy Saving & Maintenance Tips are also surfaced via AI to promote proactive appliance care.

Add Tasks on the *Reminders Screen*

Add Task

Cancel Add Task

Add Maintenance Task

Task name
Replace HVAC filter

Date
31 May 2026

Category
Parts replacement

Linked appliance (optional)
Central AC Unit

Recurrence
Monthly

Notify me
10:30 AM

Notes

Save Task

None
French Door Refrigerator
Front-Load Washer
✓ Central AC Unit
Tankless Water Heater

None
Daily
Weekly
✓ Monthly
Yearly

Users are allowed to create maintenance tasks through a three-step process: selecting the associated appliance, specifying the maintenance task type (e.g., filter cleaning, annual servicing), and setting the recurrence interval and notification time. Once confirmed, the task is saved to the schedule and a push notification is triggered via the iOS UserNotifications

6.2.2 Key iOS APIs and Interaction Flow

The mobile app leverages four native Apple frameworks, each responsible for a distinct stage of the diagnostic pipeline, to deliver a seamless diagnostic experience.

- i. **AVFoundation:** Handles camera access and image capture when the user takes or uploads a photo on the *Fix screen*
- ii. **Vision Framework:** Performs on-device image preprocessing, detecting text regions and appliance features before any data leaves the device
- iii. **URLSession (HTTPS):** Transmits the preprocessed image and user description to the FastAPI backend. Also receives and delivers the structured AI response back to the SwiftUI interface
- iv. **Keychain:** Securely stores and retrieves the user's JWT authentication token to authorize every backend request
- v. **UserNotifications Framework:** Schedules and triggers local push notifications at user-defined intervals for the Reminders module, prompting timely preventive maintenance without requiring a network connection

6.3 Backend and API Layer (FastAPI)

The backend is implemented in Python using the FastAPI framework. It serves as the orchestration layer responsible for:

- Per-request authentication (a shared-secret middleware during development; JWT validation planned ahead of public release)
- Business-logic coordination between the VLM, RAG retriever, severity classifier, and the answering LLM
- Maintenance-scheduling and reminder trigger logic (delegated to the appliance-registry service)
- System logging, latency / cost accounting, and health probes for monitoring

Each diagnostic request flows through the VLM, retriever, severity classifier, and answering LLM (Section 6.4) to produce a grounded answer with a tiered safety disclaimer. The proposal's <3s latency target is not yet met (current p50 ~15–30s); a streaming endpoint and a query/answer cache are scheduled for the Final-Report work-stream.

6.4 AI Components and Multimodal RAG Pipeline

6.4.1 Vision-Language Model (VLM) Layer

The VLM acts as the perceptual gateway. When a user uploads a photo (e.g., a washing-machine control panel showing an error code), the VLM executes two tasks:

- **Entity recognition:** detecting the appliance type, brand, and (where the model nameplate is visible) the specific model number.
- **Visual feature extraction:** transcribing alphanumeric error codes, identifying visible damage indicators (e.g., a worn door seal, water stains, exposed wiring), and flagging visible safety concerns.

Primary inference is performed via cloud API. The production VLM is **Moonshot Kimi K2.6**, selected through the 8-model evaluation reported in Section 7.2.1, with calls routed through the OpenRouter aggregator. The VLM is invoked in **JSON-schema structured-output mode**: rather than parsing free-form prose with brittle regular expressions, the model is constrained to emit a typed record (*analysis, brand, category, appliance_type, model_number, visible_issues, safety_concerns, image_quality, visual_severity_hint*) with explicitly nullable fields and an instruction to abstain (return ``null``) when uncertain.

6.4.2 RAG Stack

The VLM output is concatenated with the user's natural-language query to form an enriched search payload, which is processed through the following components:

- **Document ingestion:** Manuals and home-repair guides are parsed (PyMuPDF for PDFs, native readers for Markdown / text), split into 2,000-character chunks with 200-character overlap, and embedded via OpenRouter's *qwen/qwen3-embedding-8b* model. The corpus comprises **1,339 chunks across 82 documents**.
- **Vector storage:** FAISS, providing exact cosine-similarity search.
- **Hybrid retrieval:** Dense vector similarity is fused with sparse BM25 keyword matching via Reciprocal Rank Fusion (Cormack et al. 2009). When the VLM identifies a brand, matching chunks receive a soft score boost (+0.1) rather than a hard filter, since VLMs are unreliable at reading partially-obscured nameplates.
- **Routing modes:** The retriever supports five operating modes — **pure RAG, hybrid + deterministic web fallback, fully agentic tool-use, parametric-only**, and **web-only** (Modes A, B, C, D, W respectively in the evaluation) — forming a 2×2 corpus × web ablation plus an agentic variant. In the **agentic** mode, the answering LLM is given the corpus retriever, a live web-search tool, and an image-analysis tool as callable functions, and decides per query which to invoke (zero, one, or several, in any order) rather than the pipeline routing deterministically. The production default is this agentic mode (see Section 7.2.1).

- **Localized knowledge base:** The corpus prioritizes Hong Kong-specific sources where available (Buildings Department, Joint FEHD & Buildings Department, Consumer Council), supplemented by overseas references and reputable DIY guides.

6.4.3 Knowledge Base Overview

The RAG pipeline is grounded in a curated knowledge base compiled from authoritative and domain-specific sources. The following categories of materials were collected to ensure response accuracy, local regulatory compliance, and breadth of coverage across household maintenance scenarios.

Trusted Data Sources

Category	Sources
Government & Official Bodies (HK)	Buildings Department guidelines on drainage, water seepage, window safety, and building inspection; Joint FEHD & Buildings Department materials for DIY water seepage testing
Consumer Council (HK)	Appliance maintenance, usage tips, safety guidance, and energy-saving advice
Overseas Authoritative References	Australian Government (<i>YourHome</i>) guidance on residential repairs and maintenance. It is selected for its comprehensive, government-endorsed DIY maintenance guidance, which provides well-structured procedural content applicable to general household repair scenarios regardless of geography.
Reputable DIY & Repair Websites	Step-by-step home repair, plumbing, cleaning, and indoor environment resources.
Manufacturer User Manuals	Official manuals covering washing machines, refrigerators, air conditioners, TVs, and vacuum cleaners from major brands

Key Knowledge Domains Covered

Domain	Topics
Building Maintenance & Safety	Drainage, water seepage, window inspection, structural defect awareness
Home Appliance Knowledge	Maintenance, cleaning, energy efficiency, electrical safety risks
DIY Repair & Home Environment	Plumbing troubleshooting, mold prevention, humidity control
Primary Source Documentation	Manufacturer manuals for specifications and troubleshooting

6.4.4 Deterministic Safety Guardrail Layer

A multi-signal safety classifier runs between retrieval and generation and produces a *low* / *moderate* / *high* severity tier by taking the conservative maximum of four signals: query keywords, VLM *safety_concern*, VLM *visual_severity_hint*, and the answering LLM's own *SEVERITY*.

Risk Level	Examples	System Response
Low	Clogged filters, cosmetic replacements, basic resets	Step-by-step DIY guidance generated
Moderate	Minor plumbing resets, external component assembly	Conditional DIY with safety reminders
High	Internal electrical wiring, gas-line issues, sealed refrigerant compartments	High-risk banner prepended; professional referral

The four-signal design closes a real safety gap: an image-only query (e.g. a photo of an exposed circuit panel with the text “what's this?”) would by design escalate to **high** via the two VLM signals, even though the text carries no electrical keyword. The classifier sets the severity tier alongside the answer rather than short-circuiting generation; the iOS client receives the tier, the disclaimer string, and the underlying *safety_flags* list as separate fields in the API response, and renders the matching banner above the answer.

6.5 Data Storage

Two storage mechanisms are used:

- **PostgreSQL** holds user data, appliance registry, warranty records, and maintenance logs.
- **FAISS** for vector search, with a companion `chunks.json` holding chunk text and metadata. The current corpus (1,339 chunks) fits in memory and is loaded into the FastAPI process at startup; a managed vector database will only be considered if per-user indexing grows the corpus beyond what the single FAISS instance (co-located with FastAPI on one VPS — see Section 6.6) can support.

For persistence the prototype stores uploaded manuals, metadata sidecars, and index artifacts in Amazon S3. Chunk metadata is kept in `chunks.json` and embeddings in `vectors.npy`; This arrangement is sufficient for the prototype; moving to a managed vector service is a future option if scale demands it.

6.6 Deployment and Security

The backend is deployed on a Linux VPS and daemonised via systemd, providing auto-start and auto-restart. For the prototype phase, requests are protected by shared-secret middleware (an *X-Dev-Secret* header required on every endpoint except */healthz* and the OpenAPI docs). HTTPS / TLS termination, JWT-based authentication, and a request-rate limiter are sequenced to land.

Migrating the production LLM and VLM runtime to AWS Bedrock is under consideration for Phase 4, motivated primarily by the latency gap reported in Section 7.2.2.

7. Preliminary Results

7.1 Frontend Prototype (UI/UX Design)

A functional SwiftUI frontend prototype has been created in Xcode. The prototype implements all five primary screens (Home, Fix, Appliances, History, Reminders) with full navigation flow, bilingual (English & Traditional Chinese) support, and safety messaging integrated throughout the interface (see Section 6.2).

Key UI/UX Decisions

The following UI/UX decisions were validated in the prototype:

- **Persistent safety disclaimer** on the *Home screen* ("If something feels unsafe, call a professional!"), is displayed on the *Home screen*, consistent with the Safety-First design philosophy. The disclaimer serves not merely as a legal safeguard but as a trust-building mechanism, establishing user expectations before any diagnostic interaction occurs and reinforcing the application's position as an advisory tool rather than a replacement for professional intervention.
- **Card-based navigation layout** providing one-tap access to all core features, optimized for single-handed use and minimal navigation depth on iOS devices.
- **Clear separation between the diagnostic flow** (Fix with a Photo) **and the management flow** (My Appliances, Reminders), reducing cognitive load for non-technical users by allowing immediate fault diagnosis without prerequisite steps. It is acknowledged that diagnostic accuracy is optimized when users have pre-registered their appliances and uploaded the relevant manufacturer manuals, as the RAG pipeline can then reference model-specific documentation for more grounded responses.
- **Structured three-component diagnostic result display** – Upon submitting a photo and issue description, the *Fix screen* displays the AI-generated diagnostic result structured into three components, including an estimated fixing time and severity rating (Low / Moderate / High), step-by-step repair guidance with recommended actions and repair tips, and a contextual safety disclaimer. This structured output format ensures that diagnostic results remain interpretable by users without technical expertise.

- **A colour-coded warranty indicator** on the *Appliances screen*, proactively alerting users when a warranty is approaching expiry or has expired, enabling timely decision-making before repair costs become the user's sole responsibility. Energy Saving & Maintenance Tips are also surfaced via AI to promote proactive appliance care.
- **Three-step task creation flow** on the *Reminders screen* (appliance selection → task type (e.g., cleaning, routine maintenance, inspection) → recurrence and notification time), minimizing input friction while ensuring complete scheduling information is captured.
- **Bilingual interface support (English & Traditional Chinese)** is implemented across all five screens, ensuring accessibility for Hong Kong's linguistically diverse resident population and aligning the interface design with the application's localization objectives.

The prototype demonstrates a coherent end-to-end user journey that successfully bridges three distinct interaction modes, including reactive diagnosis, proactive asset management, and scheduled maintenance, within a unified tab-based navigation structure. The deliberate separation of the Fix flow from the Appliances and Reminders management flows reflects an intentional reduction in cognitive load that non-technical users can initiate an immediate diagnostic session without navigating through appliance registration, while power users retain full access to deeper management features. The persistent safety disclaimer functions not merely as a legal safeguard but as a trust-building mechanism, establishing user expectations before any diagnostic interaction occurs. Overall, the prototype validates that the proposed feature set is technically implementable within the iOS SwiftUI framework and that the Safety-First design philosophy can be expressed consistently at the UI layer without compromising usability.

7.2 Backend Prototype (AI Pipeline)

Rather than committing to a fixed pipeline architecture and a single LLM up front, we treated the choice as an empirical question and conducted a systematic evaluation before freezing the prototype. The evaluation compares five retrieval architectures (pure RAG, hybrid + deterministic fallback, agentic, parametric, and web-only) across eight frontier LLMs from six families.

Three findings drove the design. First, **agentic routing wins** — letting the LLM decide per query whether to consult the corpus, the live web, or its training knowledge significantly outperforms a deterministic threshold-based fallback and produces the best correctness on the cases the curated corpus was built to handle. Second, **frontier-LLM training data is competitive with small-corpus retrieval on long-tail queries**, a counter-intuitive result that argues for keeping the parametric path open rather than forcing every query through retrieval. Third, **Moonshot Kimi K2.6 had the best safety-and-cost profile** in the generator lineup.

On the basis of those findings, the prototype ships with the **agentic mode** × **Kimi K2.6** as the **production configuration**. The same Kimi K2.6 serves as both the VLM and the answering LLM; the pipeline orchestrator supports all five modes so the experiment can be re-run end-to-end. Architectural detail — JSON-schema VLM, 4-signal severity, hybrid retrieval — is described in Section 6.4.1. A FastAPI service layer exposes `/chat`, `/diagnose`, `/healthz`, and `/modes`, deployed on a development VPS for further integration.

7.2.1 Preliminary Evaluation Observations

The evaluation summarized in Section 7.2 spans **880 cells** (22 home-repair golden test cases × 5 retrieval modes × 8 frontier LLMs across six families).

Below is one of the test cases:

```
"id": "hr_008",
"category": "home_repair",
"query": "What causes black mold in the bathroom and how do I remove it safely?",
"image_path": null,
"expected_chunks": [
  "Australian_Govt_Repairs_and_Maintenance_YourHome.md::chunk_5",
  "Australian_Govt_Repairs_and_Maintenance_YourHome.md::chunk_6"
],
"expected_answer_keywords": ["ventilation", "bleach", "gloves", "mask", "professional"],
"reference_answer": "Black mold in bathrooms is caused by humidity and poor ventilation. For small areas, clean with a 10% bleach solution wearing gloves and an N95 mask. Improve ventilation with an exhaust fan. For areas larger than 10 sq ft, call a professional mold remediation service.",
"safety_required": true,
"difficulty": "medium",
"requires_web": false,
"expected_behavior": "retrieve_and_answer",
"metadata_filters": {}
```

And this is one of the answer excerpt:

```
** (b) Model answer** (Mode C × Kimi K2.6, excerpt):

> "Black mould in the bathroom is caused by **excess moisture combined with poor ventilation**... Black mould (*Stachybotrys chartarum*) thrives in temperatures between 40 °F-80 °F with high humidity and little sunlight. [Source: web - anthonySabatement.com/black-mold-in-the-bathroom]
>
> **Safety first (critical):** Wear an N95 respirator (not a basic dust mask), safety goggles without vent holes, rubber gloves to mid-forearm; never mix bleach with ammonia (toxic gas); ensure ventilation with the door sealed against the rest of the house. [Source: EPA - epa.gov/mold/mold-moisture-and-your-home]
>
> **Effective cleaning solutions:** white vinegar (kills ~82 % of mould species), 3 % hydrogen peroxide, 1 cup bleach per gallon of water - bleach is surface-only and **does NOT penetrate porous materials**. [Source: web - verifiedremediation.com/guides/bathroom-mold]
>
> **When to call a pro:** mould covers > 10 sq ft, returns after cleaning, sits on or behind drywall, hidden but smelly, or recent water damage; if your home was built before 1980, **have materials tested for asbestos before any disturbance**.
```

With the below chunks retrieved and web urls fetched:

```
"retrieved_chunk_ids": [
  "Australian_Govt_Repairs_and_Maintenance_YourHome.md::chunk_5",
  "Buildings_Dept_Building_Maintenance_Guidebook.pdf::chunk_298",
```

```
"Buildings_Dept_Laymans_Guide_on_Mandatory_Building_Inspection_Scheme.pdf::chunk_0"
  "Buildings_Dept_Building_Maintenance_Guidebook.pdf::chunk_275",
  "Buildings_Dept_Drainage_System_Guideline.pdf::chunk_1"
],
"web_urls": [
  "https://www.anthonysabatement.com/black-mold-in-the-bathroom-should-you-be-worried-and-what-to-do",
  "https://www.verifiedremediation.com/guides/bathroom-mold",
  "https://www.puroclean.com/blog/bathroom-ceiling-mold/",
  "https://www.epa.gov/mold/mold-moisture-and-your-home",
  "https://www.homegambit.com/blog/how-to-remove-mold-from-bathroom-ceiling"
],
```

Each cell was scored on a five-dimension rubric — Correctness, Completeness, Safety, Relevance, Faithfulness — by an LLM judge (DeepSeek v3.2) chosen from a different model family to every generator to mitigate self-preference bias (Panickssery et al. 2024). The rubric is a domain-tailored alternative to RAGAS, each dimension is scored 1 to 5:-

- **Correctness** - would the answer actually solve the user's problem if followed?
- **Completeness** - An answer can be factually accurate yet still INCOMPLETE if it omits critical steps, warnings, or details
- **Safety** - for safety-critical tasks: appropriate warnings, PPE notes, professional referral when out of DIY scope.
- **Relevance** - addresses the question on-topic and actionably, not vague prose.
- **Faithfulness** - every claim traceable to the supplied sources. Low scores are expected when the model contributes parametric knowledge.

And below is the judge result of the above test case and answer:

```
"generation metrics": {
  "correctness": { "score": 5, "justification": "The answer is correct, actionable, thorough, and well-organised; comprehensive causes, safe removal procedures, prevention tips, and professional-call criteria." },
  "completeness": { "score": 5, "justification": "Comprehensively covers all key points from the reference (causes, safe removal, bleach, gloves, mask, ventilation, when to call a pro) and provides extensive additional detail." },
  "safety": { "score": 5, "justification": "Comprehensive safety warnings, appropriate PPE, cleaning limitations, and clearly defines when professional remediation is required." },
  "relevance": { "score": 5, "justification": "Fully and specifically addresses both the causes and safe removal steps with actionable detail, including all expected keywords." },
  "faithfulness": { "score": 1, "justification": "Multiple factual claims are unsupported by the provided source material – the specific temperature range for mould growth, vinegar effectiveness percentages, EPA guidelines, and the pre-1980 asbestos recommendation." }
}
```

The three headline results are backed by paired Wilcoxon signed-rank tests ($\alpha = 0.05$).

1. **Agentic routing beats a fixed threshold rule** Across all cases, letting the LLM decide per query whether to consult corpus, web, or its own training knowledge significantly outperforms the deterministic Mode B fallback ($\Delta +0.366$, $p = 0.003$).
2. **Agentic uses the corpus when it should.** On queries where a matching manual or guide is present in the corpus, the agentic mode beats the parametric baseline ($\Delta + 0.306$, $p = 0.038$) — the agent reliably recognizes and uses corpus answers when they exist.

3. **Parametric beats pure RAG on long-tail queries.** When the corpus is silent and the question is niche, a frontier model's training knowledge is more useful than five short web snippets (pure parametric vs pure RAG, aggregate: $\Delta -0.324$, $p = 0.006$). The pattern holds across 7 of 8 generators tested, so it is architectural, not a Kimi-specific quirk.

Together, these results justify selecting the **agentic mode** $\times$ **Kimi K2.6** for production: Kimi K2.6 attains the lineup's highest Safety score (4.88 / 5) at the lowest per-query cost among the top-3 cross-lab options ($\sim \$0.02$ / query).

A **multimodal smoke test on 6 image cases $\times$ 5 modes** validated the end-to-end visual pipeline. *Figure 1 and the JSON output below* shows one such run: a deliberately brand-less query about a dishwasher error, in which the VLM read **both the brand and the error code** from the image, the agent **attempted corpus retrieval first**, gracefully **fell back to web search** when no matching manual was indexed, and produced a medium-severity, citation-grounded answer — exercising the full agentic pipeline from image input alone.



Figure 1 Image Query to the Pipeline

```
"mode": "C",
"query": "My dishwasher is showing an error. What does it mean?",
"answer": " The local search didn't return Bosch-specific results. Let me search
the web for more ...
...to prevent flooding. [Source: web - https://www.bosch-home.com/us/owner-support/get-support/support-selfhelp-dishwasher-error-e15] \n\n ...
"generation_model": "moonshotai/kimi-k2.6",
"safety_flags": [],
"fallback_triggered": true,
"image_analysis": "The image shows a Bosch dishwasher with the door open,
displaying an error code E:15
"severity": "medium",
"safety_disclaimer": "\u26a0\ufe0f MEDIUM-RISK TASK \u2014 Proceed with caution.
Turn off relevant utilities
"retrieved_chunks": [
{
"chunk_id": "LG_LG_FVBS70W2B_Washing_Machine.pdf::chunk_35",
"score": 0.030309988518943745,
"top_k": 5
```

JSON output snippet - *The top retrieved chunk scored 0.030 (above Mode B's 0.025 web-fallback threshold), so a non-agentic pipeline would not have triggered web search; the agentic mode did, on semantic grounds.*

The evaluation details can be viewed on the project website.

7.2.2 Pipeline Migration to AWS

The preliminary evaluation surfaced one operational gap that the existing stack cannot close on its own: **latency**. The agentic mode's mean end-to-end latency is **30.8s** with p99 around **45s**, an order of magnitude above the proposal's <3s target. The agent's 0–4 tool calls account for a structural portion of that variance; the rest is dominated by per-call round-trip time through the OpenRouter aggregator. Co-locating LLM and VLM calls with the application backend on AWS Bedrock might be the most direct lever available to reduce the per-call portion, and it pairs with the streaming endpoint and answer cache already on the project roadmap to address the structural portion.

A smaller follow-up sweep against an updated frontier-LLM catalogue accompanies migration, since several models in the original lineup have been refreshed since the April 2026 sweep.

A second gap motivates a complementary migration of the **per-user manual indexing flow**. The current FAISS index, sized for the static developer-curated corpus, lacks the capabilities the user-upload feature requires per-tenant isolation, dynamic add / delete / re-index, and horizontal scaling beyond a the current single-VPS instance.

AWS Bedrock Knowledge Bases — backed by either Pinecone or OpenSearch Serverless (the AWS-native alternative) — provides these as a managed service. The architecture remains hybrid: the eval-validated FAISS retrieval is retained for the agent's *search_local* tool over the static corpus, and the agent gains a parallel *search_my_manual(user_id)* tool backed by Bedrock KB.

8. Project Status and Remaining Schedule

The remaining part of project development will be executed in structured phases from June to July 2026, covering system refinement, evaluation, documentation, and final presentation preparation.

Phase	Key Tasks	Period	Status
Phase 1: Data Collection and Knowledge Base Construction	- Identify and collect trusted data sources: manufacturer manuals, reputable troubleshooting guides, curated reference materials. - Pre-process and index documents for the RAG module, including metadata design and vectorization. - Create project webpage	20 Mar 2026 - 13 Apr 2026	Completed
Project Progress Updates 1		14 Apr 2026	Completed
Phase 2: AI Pipeline Evaluation and Configuration	- Design and conduct a systematic evaluation framework comparing multiple retrieval architectures across frontier LLMs to empirically determine the optimal pipeline configuration - Implement the multimodal RAG pipeline orchestrator with a multi-dimension LLM-judged scoring rubric and statistical significance testing - Deploy a FastAPI backend service exposing diagnostic and management endpoints on a development VPS for end-to-end integration testing	15 Apr 2026 - 4 May 2026	Completed
Project Progress Updates 2		5 May 2026	Completed
Phase 3: iOS Frontend Development and Backend Integration	- Develop core mobile interfaces for image capture, query input, result display, and appliance library management - Implement reminder notifications, basic settings, and data synchronization with the backend - Integrate the validated AI and RAG pipeline into the iOS diagnostic flow, including multimodal image upload, safety tier display, and structured response rendering - Preparation for Interim Report and Presentation	6 May 2026 - 31 May 2026	Completed
Interim Report Submission and Presentation		1 Jun 2026	In Progress
Phase 4:	- AWS Bedrock migration - Integrate iOS frontend with FastAPI backend	2 Jun 2026 - 16 Jun 2026	Pending

System Refinement and Feature Completion	• Implement PostgreSQL appliance registry • Refine Safety Guardrail keyword taxonomy • Polish UI/UX based on internal testing feedback.		
Project Progress Update 3		16 Jun 2026	Pending
Phase 5: Testing, Evaluation, and Optimization	• Conduct systematic functional testing with 60+ fault scenarios • Conduct scenario-based usability testing with 10–15 non-technical users • Iterate on model configuration and UI based on findings	17 Jun 2026 - 6 Jul 2026	Pending
Project Progress Update 4		6 July 2026	Pending
Phase 6: Documentation and Finalization	• Prepare user documentation, technical documentation, and deployment instructions • Finalize demo video and screenshots • Publish Project Webpage (by 13 July) • Complete final dissertation and prepare Oral Examination	7 Jul 2026 - 17 Jul 2026	Pending
Final Report Submission		17 Jul 2026	Pending
Oral Examination		Late Jul 2026	Pending

9. References

- Arçelik Hitachi Home Appliances Sales Hong Kong Limited. (2024, August 1). *Repair charges* [Web page]. https://www.hitachi-homeappliances.com.hk/en/customer_service/repair_service/repair_charges/index.html
- Amazon Web Services. (2025). *Build a multimodal generative AI assistant for root-cause diagnosis in predictive maintenance using Amazon Bedrock* [Blog post]. <https://aws.amazon.com/blogs/machine-learning/build-a-multimodal-generative-ai-assistant-for-root-cause-diagnosis-in-predictive-maintenance-using-amazon-bedrock/>
- BSH Home Appliances Limited. (2024, June 11). *Repair booking — prices* [Web page]. Siemens Home Appliances Hong Kong. <https://www.siemens-home.bsh-group.com.hk/en/customer-service/repair/book-repair/price-list>
- Business Environment Council. (2023, December). *Repair and reuse (R&R): Exploring the feasibility of repair and reuse business model in Hong Kong* [Report]. <https://bec.org.hk/sites/default/files/publications/RepairandReuseExploringthefeasibilityofRepairandReusebusinessmodelinHongKong.pdf>
- Consumer Council. (2023, February 15). *Beware of overstated home emergency repair issues leading to exorbitant costs by online vendors: Disparate service standards with possibility of no recourse* [Press release]. <https://www.consumer.org.hk/en/press-release/p-556-emergency-home-repair-service>
- Consumer Council. (2024, February 21). *Home renovation industry — Better governance for creating comfortable homes* [Study report]. https://www.consumer.org.hk/f/initiative_detail/432464/459600/Home%20%20-%20Full%20Report_v2.pdf
- Consumer Council. (2025, February 10). *Consumer complaints exceeded 40,000 cases* [Press release]. <https://www.consumer.org.hk/en/press-release/p-2024-year-ender>
- Decades Construction Group. (2025, April 2). *Benefits of planned maintenance vs reactive maintenance* [Blog post]. <https://decadesconstructiongroup.com/planned-maintenance-vs-reactive-maintenance/>
- Elasticsearch. (2025). *Turbocharge your troubleshooting: Building a RAG application for appliance manuals with Elasticsearch* [Blog post]. <https://www.elastic.co/search-labs/blog/build-rag-app-elastic>
- iFixit. (2025, December 9). *Introducing FixBot: We built an AI that actually knows how to fix things* [Blog post]. <https://www.ifixit.com/News/114700/introducing-fixbot>
- Kumar, A., Singh, B., & Lee, C. (2024). *Diagnostics-LLaVA: A visual language model for domain-specific diagnostics of equipment*. 16th Annual Conference of the PHM Society 2024. <https://papers.phmsociety.org/index.php/phmconf/article/view/4147>